



Phishing Simulation Tool

Targets To Achieve:

- Simulate phishing attacks to educate students.
- Customizable email campaigns and user response tracking.
- Awareness scoring and reporting.

Simulation Stages

1. Planning and Scope Definition

Task	Description
Define Objectives	Clearly outline what participants should learn from the simulation (e.g., identify malicious links, recognize social engineering tactics).

Task	Description
Identify Target Audience	Determine the group of individuals who will participate in the simulation.
Create Realistic Phishing Scenarios	Design phishing emails that look like real ones. Some of the Ideas are: • Credential harvesting (e.g., fake login page) • Click-only (tracking clicks only) • Attachment-based (with fake files) • Spear phishing (targeted individual)
Send the Phishing Emails	Schedule the campaign: • It should be Random and not predictable. • If sending students, then possibly best ideas for their awareness are: 1. Reset your password for security reasons. 2. Your socialmedia app(like instagram or twitter) password is outdated, please reset it for updated security.
Monitor User Responses	Track and log: • Who clicked the link • Who entered credentials • Who reported the email • Who ignored it

Task	Description
	We create a dashboard where the results are shown.
Analyze the Results	Generate a report including: • Click rate • Submission rate (users who entered passwords) • Report rate (users who flagged it) • Repeat offenders This gives insight into: • Who is at risk • How many people fall for phishing • Where to focus awareness training
Provide Feedback	After the campaign: • We show users their performance and tell them that they have fallen for a trap and make them aware of such attacks happening. • Offer awareness training or video tutorials

Task	Description
	- Teach them how to: - Spot phishing - Report suspicious emails - Use multi-factor authentication (MFA)
Repeat	We can Repeat this simulation once every month depending on the situation by trying different attacks and latest technology and educate everyone from falling for these common attacks. Overtime, there will less and less people falling for these and will be ready for corporate world to face these challenges.

2.Skills Required

Task	Things to Know	Skills Required	Tools Required
To Simulate Phishing Attacks	1] Understanding of phishing techniques (fake login forms, link manipulation). 2] Frontend web development to design realistic phishing pages	HTML CSS JAVA SCRIPT - For developing a fake website where all the people who clicked the link can land in. - This website will contain all the effects and statistics of the phishing attacks and will teach them how to avoid them next time so they learn from their mistake. - Hosting fake landing pages and tracking input	Web Development JavaScript, Bootstrap (frontend), Flask/Django (Python backend)

Task	Things to Know	Skills Required	Tools Required
Customizable Email Campaigns	Customizable email campaigns allow you to: • Send realistic fake phishing emails to test users • Vary the difficulty level of the phishing attempts • Use templates (like password reset, invoice due, survey link, etc.) • Track: ◦ Who clicked the link ◦ Who entered credentials ◦ Who reported the email	• Email HTML design. • Working with SMTP(simple mail transfer protocol) servers or Email APIs. • Python scripting to automate sending emails. • Adding personalized tokens/links to each email.	<i>Tools That Support Customizable Campaigns:</i> • GoPhish • KnowBe4 • Cofense (PhishMe) • SendGrid, Zoho Campaigns, etc. These platforms allow us to: • Create email templates • Import recipient lists • Set up automation rules • Track analytics (open rate, click rate, etc.) These are the tools which we need to master to customize the emails perfectly so that most likely the link is clicked. <i>Apart from these Email template generators, we can also customize emails with:</i> • Python smtplib, email libraries • SendGrid/Mailgun APIs
User Response Tracking	We need to track the information on how many people have clicked out of all the people sent to	• JavaScript for tracking clicks/form submissions	• Flask with routes to log data • JavaScript event logging

Task	Things to Know	Skills Required	Tools Required
	educate them about their mistakes.	- Backend to record timestamp, user email, IP, browser info - Cookies or tokens to identify users uniquely	PostgreSQL/MySQL to store responses
Awareness Scoring & Reporting	Secure informed consent from all participants, clearly explaining the nature and purpose of the simulation.	- Python for analyzing user responses - Generating scores based on behavior (clicked link = -1 point, reported = +1 point, etc.) They can report about the suspicious mail to the official Cybersecurity club mail ID provided to them. By Tracking and maintaining points based system, it generates some level of competitiveness among the students so that they avoid the sketchy mails which they get. - Maintaining all the Data in forms of pie charts and bar graphs for easy understanding - Creating downloadable reports (PDF, CSV)	- Pandas, Matplotlib, Seaborn - Flask-Admin/Django Admin (dashboard) - Charts.js for frontend visualization